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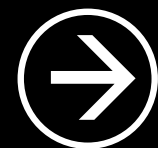


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UNIFIED AIR INTELLIGENCE PLATFORM

A shared digital system enabling anticipatory governance and informed public response in Delhi.

PROBLEM CONTEXT



- AQI frequently reaches Severe levels (400–500+) during winters
- PM2.5 often 8–12× above WHO safe limits
- Long-term exposure contributes to 1.6M+ premature deaths annually in India



Swacch Hawa
ka Saath, Sabka Prayaas

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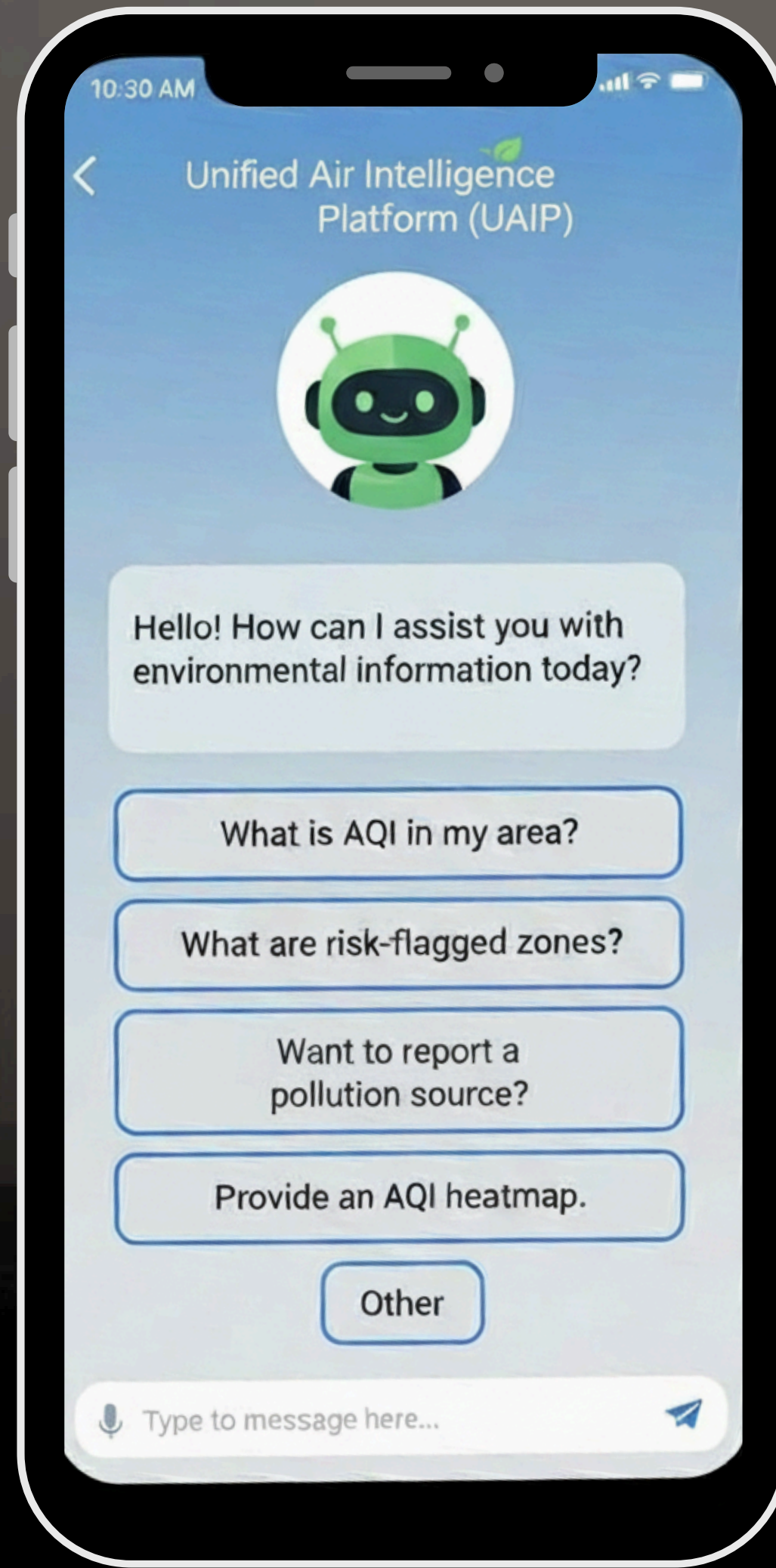


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THE SOLUTION



01

SINGLE SOURCE OF TRUTH

The bot integrates air quality, weather, traffic, and forecast data into one unified interface, eliminating fragmented information across platforms.

02

ACTION-ORIENTED INTELLIGENCE

Instead of raw AQI numbers, the bot delivers contextual insights and recommended actions for both authorities and citizens, aligned with severity levels.

03

COORDINATED COMMUNICATION

A shared bot interface ensures consistent, timely messaging to government teams and the public, enabling faster response and informed daily decisions.

DATA & INTELLIGENCE BACKBONE

MULTI-SOURCE DATA INTEGRATION

Aggregates air quality, weather, traffic, satellite, and historical data into a single, continuously updated system.

HYPER-LOCAL POLLUTION MAPPING

Uses intelligent interpolation to generate neighborhood-level air quality insights beyond fixed monitoring stations.

PREDICTIVE EARLY- WARNING MODELS

Forecasts pollution trends 24–72 hours in advance, enabling preventive action rather than reactive response.

GOVERNMENT INTERFACE: FROM ALERT TO ACTION

■ Provides a real-time city heatmap highlighting pollution hotspots, forecasted risk zones, and vulnerable locations such as schools and hospitals.

■ Converts risk thresholds into pre-approved response actions and tracks enforcement and departmental follow-through for better coordination and accountability.

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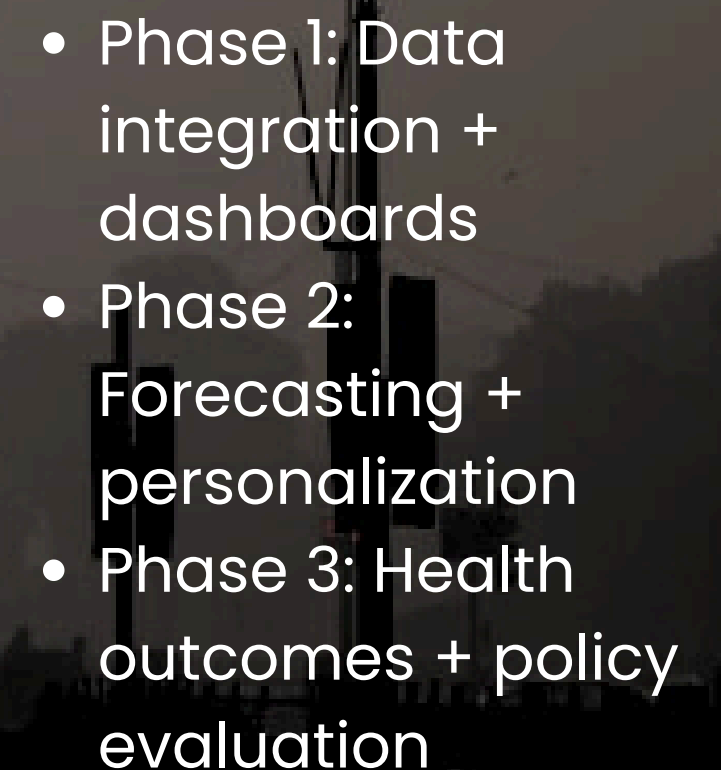
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CITIZEN INTERFACE: EXPOSURE, NOT JUST AQI

- Neighborhood-level air quality
- Actionable guidance instead of raw numbers
- Safer time windows & routes
- Optional risk profiles for vulnerable users
- Two-way reporting of local pollution sources

ROADMAP

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- Phase 1: Data integration + dashboards
 - Phase 2: Forecasting + personalization
 - Phase 3: Health outcomes + policy evaluation

OUR TEAM

VoltEdge is a multidisciplinary team of engineering students from MNNIT Allahabad, focused on structured problem-solving and clear decision-making. We identify overlooked real-world challenges and translate them into practical, scalable solutions through rigorous analysis and thoughtful design.

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Praveen



Muhammad



Mujahid

